CHAPTER 12

Geometry

12.1 Introduction

I think geometry is all about visualization and creativity. By connecting some dots or extending some line segments or rotating some angles you can easily figure out the correct answers. Practically, you need not be an artist or a painter, but at least you must be capable of visualizing the geometrical figures. So before going for the tedious and complex geometrical problems you must make yourself familiar and comfortable with the fundamental concepts of geometry. Basically, geometry consists of a very large proportion of problems in CAT. Sometimes, 15-20% problems related to this chapter are asked in CAT. Apart from CAT other competitive exams such as XAT, IIFT and SSC also ask a plethora of questions from this chapter. Therefore, it is strongly advised that you must learn this chapter religiously for the sake of a decent score.

Mind me if you don't understand the theorems, axioms or concepts minutely you cannot perform well, as the intricacy of diagrams and concepts makes it a daunting task for you.

For the sake of convenience this chapter is divided into 5 parts

1. Lines and Angles
2. Triangles
3. Quadrilaterals
4. Polygons
5. Circles and Loci

Chapter Checklist

- *Lines and Angles*
- *Triangles*
- *Quadrilaterals*
- *Polygons*
- *Circles*
- *Tangential Quadrilateral*
- ***CAT Test***

12.2 Lines and Angles

Point : The figure of which length, breadth and height cannot be measured is called a point. It is infinitesimal.

Line : A line is made up of a infinite number of points and it has only length *i.e.,* it does not has any thickness (or width). A line is endless *i.e.,* it can be extended in both directions.

Line segment : A line segments has two end points, but generally speaking line segment is called a line.

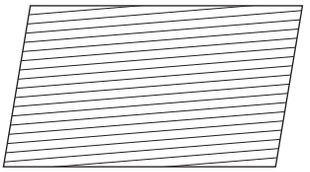


(line segment $\overrightarrow{AB}$)

Ray : A ray extends indefinitely in one direction from any given point. This is exhibited by an arrow. The starting point is called as the initial point.



Plane : It is a flat surface, having length and breadth both, but no thickness. It is a two dimensional figure.



(Plane)

Type of Lines	Definition	Diagram
Parallel lines	Two lines, lying in a plane and having no common intersecting point are called parallel lines. The distance between two parallel lines is constant.	Symbol : $\parallel$
Perpendicular lines	Two lines, which lie in a single plane and intersect eachother at right angle, are called perpendicular lines.	Symbol : $\perp$
Concurrent lines	More than two straight lines intersecting at the same point.	

Points to Remember

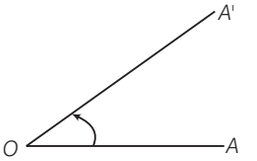
1. A line contains infinitely many points.
2. The intersection of two different lines is a point.
3. Through a given point, there pass an infinite number of lines and these lines are called **concurrent lines.**
4. Only one line can pass through any two particular points.
5. When more than two points lie on a line, they are called as **collinear points** else they are called as **non-collinear** points.
6. Two lines can intersect maximum at one point. This point is called as **point of intersection** and these lines are called as **intersecting lines**.
7. There are an infinite number of planes which pass through a single (particular) point.
8. When more than three points lie in the same plane, they are called as **coplanar,** else they are called as **non-coplanar.**
9. When more than one line lie in the same plane, then these lines are called as coplanar else they are called as non-coplanar.
10. When two planes intersect each other, they form a line *i.e.,* intersecting region is a line.
11. Two different lines which are perpendicular to the same (a third line) line, are necessarily parallel to each other, if all of them are lying on the same plane.
12. When two or more parallel lines are intercepted by some other intercepting lines, then the ratio of corresponding intercepts are equal. An intercepting line is generally called as a transversal.



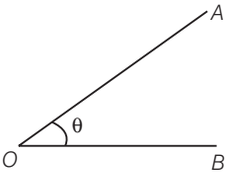
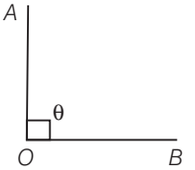
i.e., $\frac{AB}{BC} = \frac{DE}{EF}$

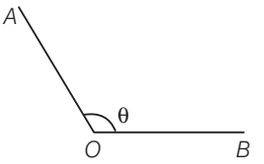
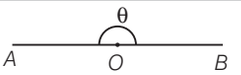
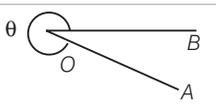
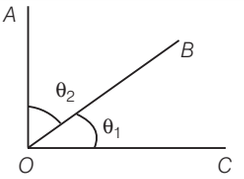
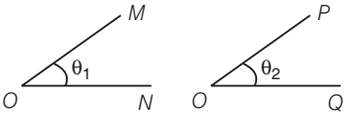
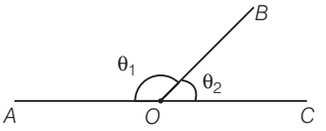
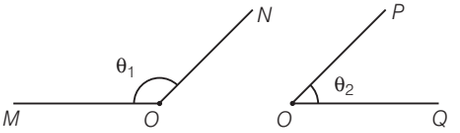
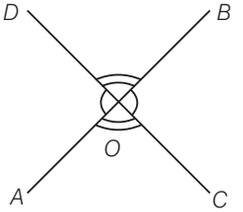
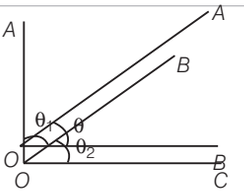
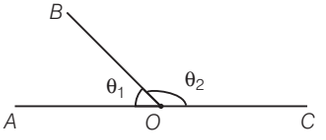
Angles

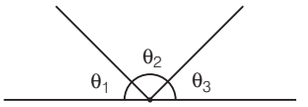
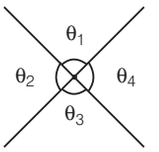
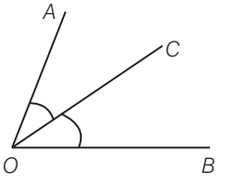
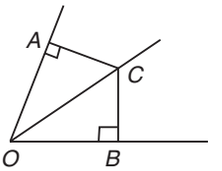
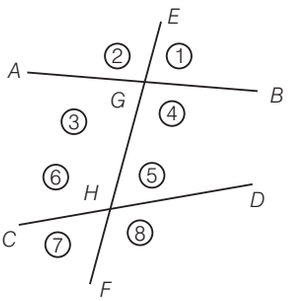
The amount of rotation about O, the vertex of the angle AOA', is called the magnitude of the angle.



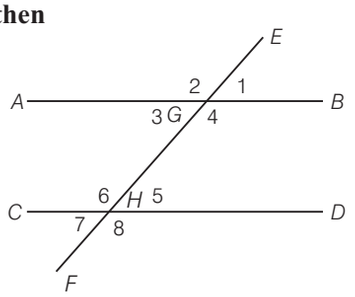
$m\angle AOA'$ denotes the measure of $\angle AOA'$. Angles are said to be congruent when their measure is same.
(symbol : $\cong$)

Types of Angles	Property	Diagram
Acute	$0° < \theta < 90°$ ($\angle AOB$ is an acute angle)	
Right	$\theta = 90°$ ($\angle AOB$ is a right angle)	

Types of Angles	Property	Diagram
Obtuse	$90° < \theta < 180°$ ($\angle AOB$ is an obtuse angle)	
Straight	$\theta = 180°$ ($\angle AOB$ is a straight angle)	
Reflex	$180° < \theta < 360°$ ($\angle AOB$ is a reflex angle)	
Complementary	$\theta_1 + \theta_2 = 90°$ Two angles whose sum is 90°, are complementary to each other	 
Supplementary	$\theta_1 + \theta_2 = 180°$ Two angles whose sum is 180°, are supplementary to each other.	 
Vertically opposite	$\angle DOA = \angle BOC$ and $\angle DOB = \angle AOC$	
Adjacent angles	$\angle AOB$ and $\angle BOC$ are adjacent angles Adjacent angles must have a common side. (*e.g., OB*)	
Linear pair	$\angle AOB$ and $\angle BOC$ are linear pair angles. One side must be common (*e.g., OB*) and these two angles must be supplementary.	

Types of Angles	Property	Diagram
Angles on the one side of a line	$\theta_1 + \theta_2 + \theta_3 = 180°$	
Angles round the point	$\theta_1 + \theta_2 + \theta_3 + \theta_4 = 360°$	
Angle bisector	*OC* is the angle bisector of $\angle AOB$. *i.e.,* $\angle AOC = \angle BOC = \frac{1}{2}(\angle AOB)$ When a line segment divides an angle equally into two parts, then it is said to be the angle bisector (*OC*)	 (Angle bisector is equidistant from the two sides of the angle) *i.e.,* $AC = BC$ 
Corresponding angles	When two lines are intersected by a transversal, then they form four pairs of corresponding angles (a) $\angle AGE, \angle CHG \Rightarrow (\angle 2, \angle 6)$ (b) $\angle AGH, \angle CHF \Rightarrow (\angle 3, \angle 7)$ (c) $\angle EGB, \angle GHD \Rightarrow (\angle 1, \angle 5)$ (d) $\angle BGH, \angle DHF \Rightarrow (\angle 4, \angle 8)$	
Exterior angles	These are following four angles (i) $\angle AGE \Rightarrow \angle 2$ (ii) $\angle CHF \Rightarrow \angle 7$ (iii) $\angle EGB \Rightarrow \angle 1$ (iv) $\angle DHF \Rightarrow \angle 8$	
Interior angles	These are following four angles (i) $\angle AGH \Rightarrow \angle 3$ (ii) $\angle GHC \Rightarrow \angle 6$ (iii) $\angle BGH \Rightarrow \angle 4$ (iv) $\angle DHF \Rightarrow \angle 5$	
Alternate angles	These are two pairs angles as following: (i) $\angle AGH, \angle GHD$ $(\angle 3, \angle 5)$ (ii) $\angle GHC, \angle BGH$ $(\angle 6, \angle 4)$	

When two parallel lines are intersected by a transversal, then



1. The pairs of corresponding angles so formed are congruent.
   i.e., $\angle 2 = \angle 6$

$$\angle 3 = \angle 7$$
$$\angle 1 = \angle 5$$
$$\angle 4 = \angle 8$$

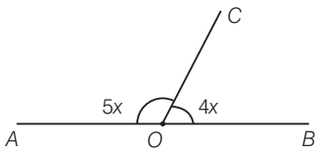
NOTE 1. If one pair of corresponding angles is congruent, then the rest pairs of corresponding angles are also congruent.

2. The pairs of alternate angles so formed are congruent
   i.e., $\angle 3 = \angle 5$ and $\angle 4 = \angle 6$
3. The pair of interior angles (*i.e.,* the interior angles on the same side of a transversal) are supplementary.

NOTE The converse of all the 3 rules is also true *i.e.,* if the pair of corresponding angles, or alternate angles or interior angles is equal (*i.e.,* congruent), then the two lines are parallel which are intersected by a transversal.

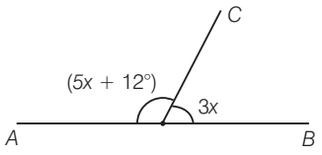
Introductory *Exercise 12.1*

1. What is the value of x in the following figure?



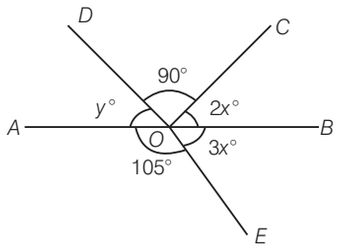
(a) 80° (b) 40°
(c) 20° (d) 25°

2. What is the value of x in the following figure?



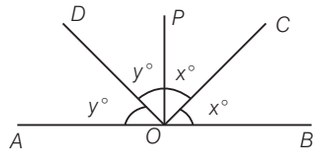
(a) 18° (b) 20°
(c) 21° (d) 24°

3. In the following figure AB is a straight line. Find $(x + y)$.



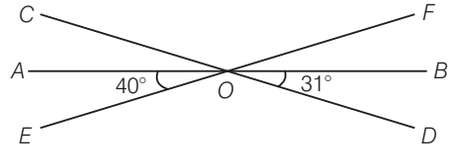
(a) 55° (b) 65°
(c) 75° (d) 80°

4. In the following figure $\angle BOP = 2x°$, $\angle AOP = 2y°$, OC and OD are angle bisectors of $\angle BOP$ and $\angle AOP$ respectively.



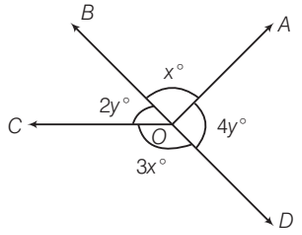
Find the value of $\angle COD$.
(a) 75° (b) 90° (c) 100° (d) 120°

5. In the following figure find the value of $\angle BOC$.



(a) 101° (b) 149° (c) 71° (d) 140°

6. Find y, if $x° = 36°$, as per the given diagram.



(a) 36° (b) 16° (c) 12° (d) 42°

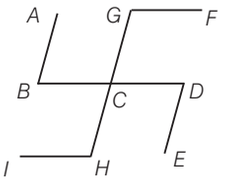
7. If $(2x + 17)°$, $(x + 4)°$ are complementary, find x.
(a) 63° (b) 53° (c) 35° (d) 23°

8. If $(5y + 62)°$, $(22° + y)$ are supplementary, find y.
(a) 16° (b) 32° (c) 8° (d) 1°

9. If two supplementary angles are in the ratio 13 : 5. find the greater angle.
(a) 130° (b) 65° (c) 230° (d) 28°

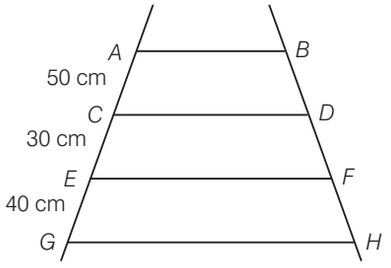
10. An angle is 30° more than one half of its complement. Find the angle in degrees.
(a) 60° (b) 50° (c) 45° (d) 80°

11. In the given diagram $AB \| GH \| DE$ and $GF \| BD \| HI$, $\angle FGC = 80°$. Find the value of $\angle CHI$.



(a) 80° (b) 120°
(c) 100° (d) 160°

12. In the given figure $AB \| CD \| EF \| GH$ and $BH = 100$ cm. Find the value of DF.



(a) 26 cm (b) 40 cm
(c) 25 cm (d) 24 cm

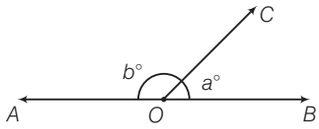
13. The complement of 65°50′ is :
(a) 24° 50′ (b) 24°10′
(c) 14° 50′ (d) 34° 10′

14. The supplement of 123° 45′ is :
(a) 56° 55′
(b) 56° 15′
(c) 55° 56′
(d) None of the above

15. If two angles are complementary to each other, then each angle is :
(a) a right angle
(b) a supplementary angle
(c) an obtuse angle
(d) an acute angle

16. How many degrees are there in an angle which equals one-fifth of its supplement?
(a) 15° (b) 30°
(c) 75° (d) 150°

17. In the given figure, $\angle a$ is greater than one-sixth of right angle, then.



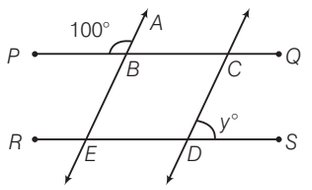
(a) $b > 165°$ (b) $b < 165°$
(c) $b \le 165°$ (d) $b \ge 165°$

18. Let D be the mid–point of a straight line AB and let C be a point different from D such that $AC = BC$, then.
(a) $AC \perp AB$ (b) $\angle BDC = 90°$
(c) $\angle BDC$ is acute (d) $\angle BDC > 90°$

19. Answer on the basis of the following statements.
When two straight lines intersect, then :
1. adjacent angles are complementary
2. adjacent angles are supplementary
3. opposite angles are equal
4. opposite angles are supplementary
(a) 2 and 3 are correct
(b) 1 and 4 are correct
(c) 1 and 3 are correct
(d) 2 and 4 are correct

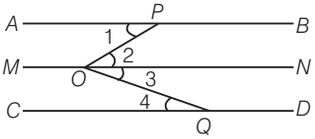
20. AB is a straight line and O is a point on AB, if a line OC is drawn not coinciding with OA or OB, then $\angle AOC$ and $\angle BOC$ are
(a) equal
(b) complementary
(c) supplementary
(d) together equal to 100

21. In the adjoining figure $AE \| CD$ and $BC \| ED$, then find Y.



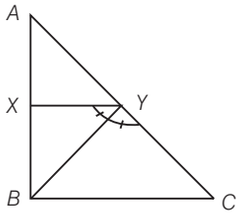
(a) 60° (b) 80°
(c) 90° (d) 75°

22. In the adjoining figure $\angle APO = 42°$ and $\angle CQO = 38°$. Find the value of $\angle POQ$.



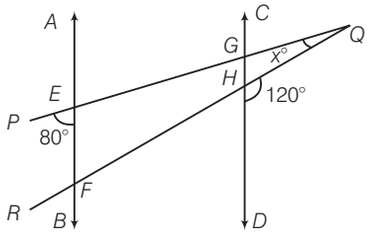
(a) 68° (b) 72°
(c) 80° (d) 126°

23. In a ΔABC, a line XY parallel to BC intersects AB at X and AC at Y :
If BY bisects $\angle XYC$, then $m\angle CBY : m\angle CYB$ is :



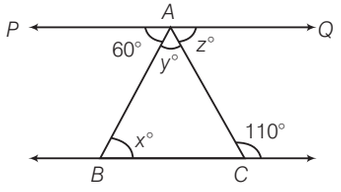
(a) 5 : 4 (b) 4 : 5
(c) 1 : 1 (d) 6 : 5

24. In the adjoining figure $AB \| CD$ and PQ, QR intersects AB and CD both at E, F and G, H respectively. Given that $m\angle PEB = 80°$, $m\angle QHD = 120°$ and $m\angle PQR = x°$, find the value of x.



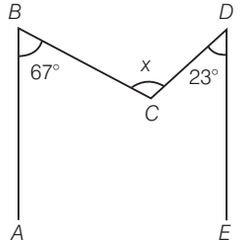
(a) 40° (b) 20°
(c) 100° (d) 30°

25. In the following figure, find the value of y.



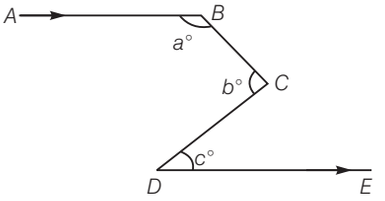
(a) 70° (b) 60°
(c) 50° (d) 80°

26. In the adjoining figure $AB \| DE$, $\angle ABC = 67°$ and $\angle EDC = 23°$. Find $\angle BCD$.



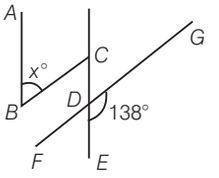
(a) 90°
(b) 44°
(c) 46°
(d) none of the above

27. In the given figure $AB \| DE$. Find $a° + b° - c°$:



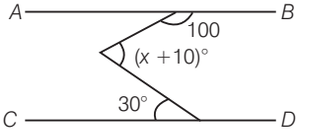
(a) 160° (b) 120° (c) 180° (d) 210°

28. In the given figure $AB \| CE$ and $BC \| FG$. Find the value of $x°$.



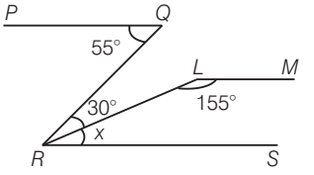
(a) 52° b) 32° (c) 42° (d) 36°

29. $AB \| CD$, shown in the figure. Find the value of x.



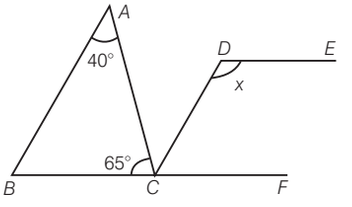
(a) 100° (b) 90° (c) 110° (d) 140°

30. In the figure $PQ \| LM \| RS$. Find the value of $\angle LRS$.



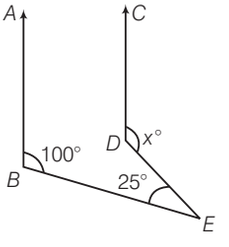
(a) 30° (b) 25° (c) 35° (d) 40°

31. In the figure $AB \| DC$ and $DE \| BF$. Find the value of x.



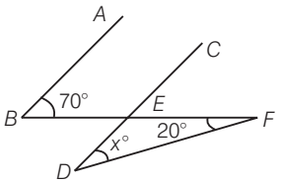
(a) 140° (b) 155° (c) 105° (d) 115°

32. In the figure $AB \| CD$, $\angle ABE = 100$. Find $m\angle CDE$.



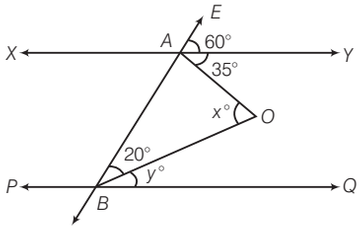
(a) 125° (b) 55° (c) 65° (d) 75°

33. In the figure $AB \| CD$, find $x°$ (*i.e.*, $\angle CDF$).



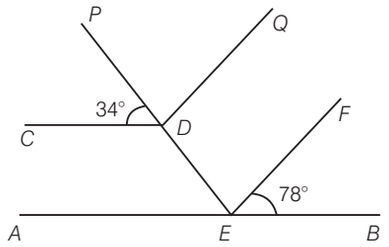
(a) 50° (b) 90° (c) 30° (d) 70°

34. In the given figure $XY \| PQ$, find the value of x.



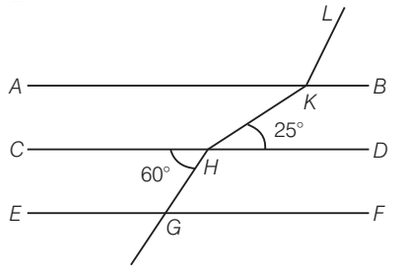
(a) 70° (b) 40° (c) 75° (d) 15°

35. In the given figure $AB \| CD$ and $EF \| DQ$. Find the value of $\angle DEF$.



(a) 68° (b) 78° (c) 34° (d) 39°

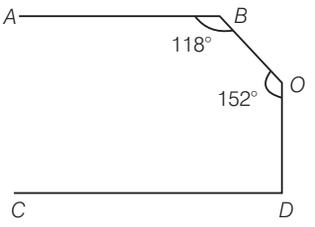
36. In the given figure $AB \| CD \| EF$ and $GH \| KL$. Find $m\angle HKL$.



(a) 85° (b) 145° (c) 120° (d) 95°

37. Which one of the following statements is false?
(a) Two straight lines can intersect at only one point.
(b) Through a given point, only one straight line can be drawn.
(c) A line segment can be produced to any desired length.
(d) Through two given points, it is possible to draw one and only one straight line.

38. $AB \| CD$. $\angle ABO = 118°$, $\angle BOD = 152°$, find $\angle ODC$.



(a) 70° (b) 80°
(c) 90° (d) 34°

39. Two parallel lines AB and CD are intersected by a transversal EF at M and N respectively. The lines MP and NP are the bisectors of interior angles $\angle BMN$ and $\angle DNM$ on the same side of the transversal. Then $\angle MPN$ is equal to :
(a) 60° (b) 90°
(c) 45° (d) 120°

40. If the arms of one angle are respectively parallel to the arms of another angle, then the two angles are :
(a) either equal or supplementary
(b) neither equal not supplementary
(c) equal but not supplementary
(d) not equal but supplementary

41. AB and CD are two parallel lines. PQ cuts AB and CD at E and F respectively. EL is the bisector of $\angle BEF$. If $\angle LEB = 35°$, then $\angle CFQ$ will be :
(a) 70° (b) 55°
(c) 110° (d) 125°

42. A plane figure is bounded by straight lines only. If n is the number of these lines, then the least value of n is :
(a) 1 (b) 2
(c) 3 (d) 4

43. How many planes can pass through any three arbitrary points?
(a) 1 (b) 2
(c) 3 (d) 0

44. Out of the four arbitrary non–collinear points three points are taken at a time, then the number of planes that can be drawn through the three points is :
(a) 3 (b) 4
(c) 6 (d) 12

45. Maximum number of points of intersection of four lines on a plane is :
(a) 4 (b) 6
(c) 8 (d) 5

Hints & Solutions

☑ Introductory Exercise 12.1

1 $4x + 5x = 180° \Rightarrow x = 20°$
($4x$ and $5x$ are linear pair angles)

2 $(5x + 12) + 3x = 180° \Rightarrow x = 21°$

3 $105 + 3x = 180 \Rightarrow x = 25$
and $y + 90 + 2x = 180° \Rightarrow y = 40°$
$\therefore x + y = 65°$

4 $2x + 2y = 180 \Rightarrow x + y = 90°$
$\therefore \angle COD = 90°$

5 $\angle DOE = 180 - (40 + 31) = 109°$
and $\angle BOF = 40°$ $(\because \angle AOE = \angle BOF)$
$\therefore \angle BOC = \angle BOF + \angle FOC$
$= 40 + 109$ $(\because \angle FOC = \angle DOE)$
$= 149°$

6 $2y + 3x = 180 \Rightarrow y = 36°$ $(\because x = 36°)$
or $x + 4y = 180° \Rightarrow y = 36°$ $(\because x = 36°)$

7 $(2x + 17) + (x + 4) = 90°$
$\Rightarrow 3x + 21 = 90° \Rightarrow x = 23°$

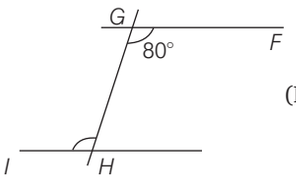
8 $(5y + 62)° + (22° + y) = 180$
$\Rightarrow 6y + 84 = 180 \Rightarrow y = 16°$

9 $13x + 5x = 180° \Rightarrow x = 10°$
$\therefore 13x = 130°$

10 Let the angle be x, then its complementary angle be $(90 - x)$
$\therefore x - \frac{(90 - x)}{2} = 30 \Rightarrow x = 50$

Alternatively The best way is to solve through options.

11



(Pair of alternate angles)

12 $\frac{AG}{BH} = \frac{CE}{DF}$
$\frac{120}{100} = \frac{30}{DF} \Rightarrow DF = 25\,\text{cm}$

13 $65°50' + x = 90° = 89° 60'$ $(1° = 60')$
$\Rightarrow x = 89°60' - 65°50' = 24° 10'$

14 $123°45' + x = 180° = 179°60'$ $(1° = 60')$
$\Rightarrow x = 179° 60' - 123° 45'$
$\Rightarrow x = 56° 15'$

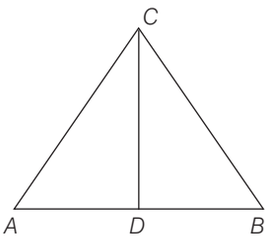
15 Since $x > 0$ and $y > 0$
Also, $x + y = 90°$
Therefore, $0 < x < 90$ and $0 < y < 90°$

16 Let the angle be x, then its supplement be $(180° - x)$.
$\therefore x = \frac{1}{5}(180° - x) \Rightarrow x = 30°$

Alternatively Go through options.
$30 = \frac{1}{5}(180° - 30°)$

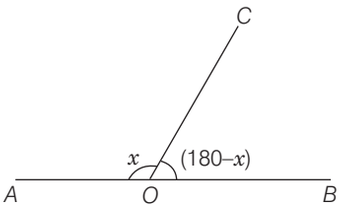
17 $a > \frac{90°}{6} \Rightarrow a > 15°$
$\because a + b = 180°$
$\therefore b < 165°$ $(\because a > 15°)$

18 $\because AC = BC$ and $AD = BD$



and CD is common
$\therefore \Delta CDA$ is similar to ΔCDB and AB is a straight line
$\therefore \angle CDA = \angle CDB = 90°$

20



21 $\angle ABP = \angle CDR = 100°$
$\therefore y = 80°$ $(\because \angle CDR + \angle CDS = 180°)$

22 $\angle APO = 42°$ and $\angle CQO = 38°$
$\angle POQ = \angle PON + \angle NOQ$
$= \angle APO + \angle OQC = 42° + 38° = 80°$

23 $\angle XYB = \angle YBC$ and $\angle XYB = \angle BYC$
$\Rightarrow \angle YBC = \angle BYC$

24 $\angle PGH = 80° \Rightarrow \angle QGH = 100°$
$\angle QHD = 120° \Rightarrow \angle CHQ = 60°$
$\therefore \angle x + 100 + 60 = 180° \Rightarrow x = 20°$

25 $\angle z = 180° - 110 = 70°$ $(\because \angle QAC = \angle BCA)$

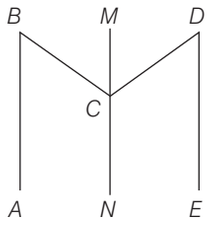
and $60° + y° + z° = 180°$

$\therefore$ $y = 180° - (70° + 60°)$

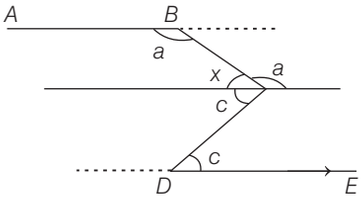
$y = 50°$

26 $\angle BCD = \angle BCM + \angle DCM$

$= \angle ABC + \angle EDC = 67° + 23° = 90°$



27 $a + x = 180°$; $c + x = b \Rightarrow x = b - c$



$\therefore$ $a + (b - c) = 180° \Rightarrow a + b - c = 180°$

28 $\angle x = \angle ABC = \angle CDG = 180° - 138° = 42°$

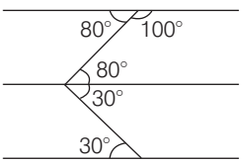
29 Use the same method as in question number 27.

i.e., $(a + b - c) = 180°$

$\Rightarrow$ $100° + (x + 10°) - 30° = 180°$

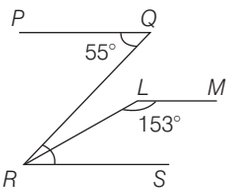
$\Rightarrow$ $x = 100°$

Alternatively



$\therefore$ $x + 10° = 80° + 30° \Rightarrow x = 100°$

30



$\angle PQR = \angle QRS$

$55° = \angle QRS \Rightarrow 55° = \angle QRL + \angle LRS$

$55° = 30° + \angle LRS \Rightarrow \angle LRS = 25°$

31 $\angle ABC = 180° - (40° + 65°)$

$\angle ABC = 75°$ and $\angle DCF = \angle ABC = 75°$

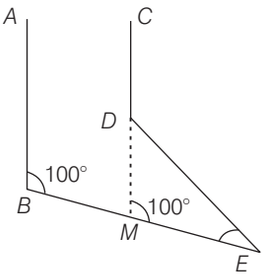
$\therefore$ $\angle EDC = 180° - \angle DCF = 105°$

32 Extend CD to M, then

$\angle DME = \angle ABE = 100° \Rightarrow \angle MED = 25°$

$\therefore$ $\angle MDE = 180° - (100° + 25°) = 55°$

$\therefore$ $\angle CDE = 180° - 55° = 125°$



33 $\angle CEF = \angle ABF = 70°$

$\therefore$ $\angle FED = 180° - 70° = 110°$

$\angle EDF + \angle DFE + \angle FED = 180°$

$\therefore$ $\angle FDE = 180° - (110° + 20°)$

$= 50°$

34 $\angle BAO = 180° - (60° + 35°) = 85°$

$\angle AOB = 180° - (85° + 20°) = 75°$

35 $\angle AEP = \angle CDP = 34°$

$\angle AEP + \angle BEF + \angle DEF = 180°$

$34° + 78° + \angle DEF = 180°$

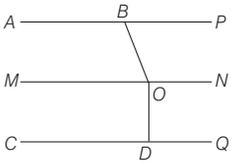
$\Rightarrow$ $\angle DEF = 68°$

36 $\angle AKH = \angle KHD = 25°$

$\angle EGH = \angle AKL = 180° - 60° = 120°$

$\therefore$ $\angle HKL = 120° + 25° = 145°$

38



$\angle ABO = 118°$

$\therefore$ $MOB = 180° - 118° = 62°$

$\therefore$ $\angle MOD = 152° - 62° = 90°$

$\therefore$ $\angle ODC = 180° - 90° = 90°$

39 Let $\angle BMN = 2x$

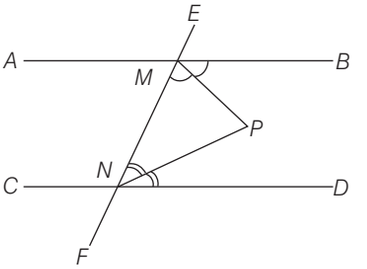
and $\angle DNM = 2y$

$\therefore$ $2x + 2y = 180°$

$\Rightarrow$ $x + y = 90°$

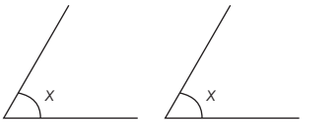
$\therefore \angle MPN = 180° - 90°$

$= 90°$



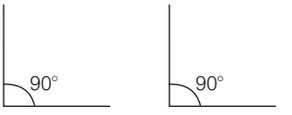
40

Equal



$x \neq 90°$

Equal and Supplementary

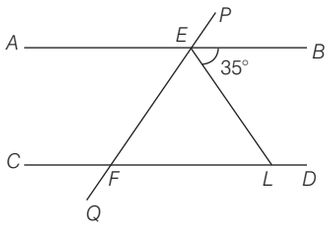


$x = 90°$

41 $\angle BEF = 2\angle BEL = 70°$

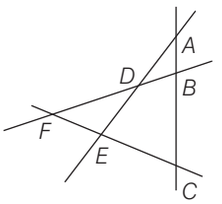
$\therefore \quad \angle CFQ = \angle EFL = 180° - 70°$

$= 110°$



44 ${}^4C_3 = \dfrac{4!}{3!1!} = \dfrac{4\cdot 3\cdot 2\cdot 1}{3\cdot 2\cdot 1\cdot 1} = 4$

45 ${}^4C_2 = \dfrac{4!}{2!2!} = \dfrac{4\cdot 3\cdot 2\cdot 1}{(2\cdot 1)\cdot(2\cdot 1)} = 6$



☑ Introductory Exercise 12.2

1 The sum of the smaller sides cannot be equal to or less than the largest side.

2 $12^2 > 8^2 + 6^2 \quad \Rightarrow$ obtuse triangle

3 Sum of exterior angles = 360°

4 $\angle ACD = \angle CBA + \angle BAC = 70° + 40° = 110°$

5 $\angle ABC + \angle ACB < 90°$

6 $x + 4x + 7x = 180°$

$\Rightarrow \quad x = 15°$

$\therefore \quad 7x = 105°$

7 $\angle OBC + \angle OCB = \dfrac{70}{2} + \dfrac{30}{2} = 50°$

$\therefore \quad \angle BOC = 180° - 50° = 130°$

8 $\angle BFC = 90° - \dfrac{1}{2}(\angle A)$

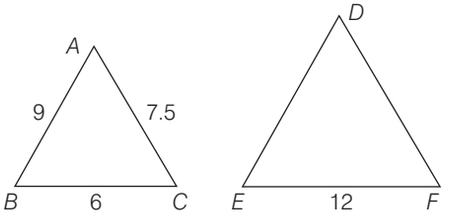
$= 90° - \dfrac{1}{2}[180° - (80° + 30°)]$

$= 55°$

10 Prove with the help of similarity.

11 Prove with the help of the similarity.

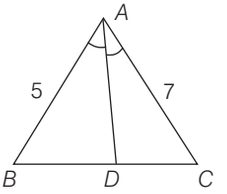
12 $\Delta ABC \sim \Delta DEF$



$\therefore \quad \dfrac{AB}{BC} = \dfrac{DE}{EF}$

$\Rightarrow \quad \dfrac{9}{6} = \dfrac{DE}{12} \Rightarrow DE = 18$ cm

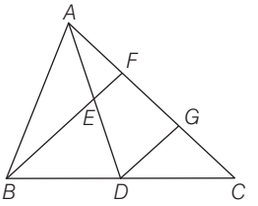
13 $\dfrac{AB}{AC} = \dfrac{BD}{CD} = \dfrac{5}{7}$ (By angle bisector theorem)



14 Let's draw a line segment DG, which is parallel to the line segment BF, such that G lies on AC.

Now look at the ΔADG, where $EF \parallel DG$, so we have

$$\frac{AF}{FG} = \frac{AE}{DE} = \frac{1}{1}$$



Now, let's have a look at the ΔBFC, where $BF \parallel DG$, so we have

$$\frac{FG}{GC} = \frac{BD}{BC} = \frac{1}{1}$$

Thus we can say that $AF : FG : GC = 1 : 1 : 1$

Therefore, $AF : FC = 1 : 2$

15 AB is hypotenuse and AD is perpendicular in ΔABD,

$\therefore \quad AB > AD$

